

Motivation letter

Dear Prof. Inge Huitinga, Prof. Joost Verhaagen, Dr. Niamh McNamara, and the Selection committee,

My name is Varvara Lazarenko, and I would like to express my strong interest in the PhD position focused on the mechanisms of myelin repair in multiple sclerosis (MS) at the Netherlands Institute for Neuroscience. With a solid academic foundation in Biology (BSc) and Medical Biology (MSc), combined with practical research experience, I bring a multidisciplinary background in neuroscience, molecular biology, and *in vivo* experimentation. I am highly motivated to contribute to this exciting project and confident that my research mindset and technical skills make me a strong fit for this position.

Throughout my education, I have received an extensive academic foundation and training in laboratory techniques central to molecular biology, genome analysis, and physiology. During my Bachelor's at Lomonosov Moscow State University, I built a robust skillset in wet lab methods, including PCR, gel electrophoresis, DNA/RNA extraction, qPCR, ELISA, immunohistochemistry, intracellular recordings, and behavioural testing in animal models. My practical experience in physiology and molecular biology was deepened during my year-long role as a Junior Research Assistant at Lomonosov MSU. There, I investigated the role of acid-sensitive two-pore potassium channels (TASK-1) in the vascular tone in rat arteries under varying pH conditions. This involved wire myography, RNA extraction, qPCR, and statistical analysis. Combined with my experience at the Contract Research Organization, this experience trained me to work independently in the lab, collaborate across interdisciplinary teams, and communicate findings through both publication and presentation, while also strengthening my documentation and project management skills.

Motivated to gain international experience, I was awarded an Orange Tulip scholarship to study a Master's in Medical Biology at Radboud University. As a part of this program, I completed a 6-month research internship at the Donders Institute, where I performed *in vivo* calcium imaging and data analysis using Python and DeepLabCut in a study of social fear learning in mice. This hands-on exposure to functional neurobiology further fueled my interest in neuroscience. Later, as a part of my Master's, I also wrote a review article on the role of acetylcholinesterase-derived peptide in Alzheimer's disease, supervised by Dr. Corette Wierenga, Professor of Neurophysiology at Radboud University.

What excites me the most about this PhD opportunity is the chance to explore such a complex disorder as MS using a multifaceted approach and a holistic perspective. The integration of bioinformatics and immunohistochemistry with *ex vivo* and *in vivo* experimentation, alongside gene therapy approaches, makes this project both innovative and highly impactful. Studying the inhibitory mechanisms of remyelination and contributing to therapeutic strategies in MS aligns deeply with my academic interests and long-standing motivation to contribute to neurodegenerative disease research.

My research experience, collaborative spirit, and deep enthusiasm for neuroscience and regenerative medicine make me a strong candidate for this position. I am excited by the opportunity to work within the dynamic, interdisciplinary environment at NIN and to further develop my scientific skills under the mentorship of the Neuroimmunology and Neuroregeneration groups.

Thank you for considering my application. Should any questions arise, please do not hesitate to contact me.

Warm regards,
Varvara Lazarenko

**Letter of Recommendation for Ms Varvara Lazarenko
to Whom It May Concern**

Arie Kim, Ph.D.
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July 8, 2025

It is with great pleasure that I write this letter of reference for Ms. Varvara Lazarenko, who is applying for a PhD position in mechanisms of myelin repair at the Netherlands Institute for Neuroscience in Amsterdam. I had the opportunity to supervise Varvara during her research internship in 2023 at the Battaglia Lab (Neuronal Networks of Memory) at the Donders Institute, where I worked as a postdoc researcher, and I can confidently speak to her abilities, motivation, and suitability for this position.

During her time in the laboratory, Varvara demonstrated a high level of scientific curiosity, technical skill, and intellectual independence. She quickly obtained the art. 9 certification required for working with laboratory animals in the Netherlands and immersed herself in our complex *in vivo* experimental setting involving two-photon microscopy. She not only supported routine animal care and feeding but also successfully practiced intricate surgical technique such as mouse craniotomy and assisted with head-fixating the animals in the behavioral setup.

Her work was part of a larger project aimed at understanding how cortical neuronal activity changes in response to different environmental sensory contexts. In our experiments, the calcium activity in cortical neurons was recorded via two-photon microscopy while mice performed a virtual reality behavioral task: head-fixed, the animals ran on a wheel while being exposed to three different sound-associated environments projected on surrounding screens. Varvara helped characterize cortical network activity by performing data analysis of neuronal recordings via Python, which she learned from scratch. She also developed a base model for tracking pupil size changes using a deep learning-based tool called DeepLabCut, which was critical for linking such psychological state markers with neuronal activity. Her analytical approach and ability to adapt to new tools and methods underscored her strong potential as a researcher.

Varvara also actively engaged in our weekly journal clubs, presenting and discussing a challenging paper related to her research. She was able to effectively communicate complex topics and participated in discussions with insights and critical perspectives, reflecting her developing academic maturity and enthusiasm for neuroscience.

On a personal level, Varvara is a highly motivated and optimistic individual. She shows resilience, strong work ethic, and a genuine passion for research. Her wide-ranging knowledge in the biomedical field, hands-on lab experience, and interest in neuroimmunology make her exceptionally well-suited for the PhD project focusing on remyelination in multiple sclerosis.

In summary, I strongly recommend Varvara Lazarenko for this PhD position. She possesses the intellectual drive, practical skills, and collaborative spirit that are essential for success in academic research. I am confident she will make significant contributions to your team and thrive in your stimulating research environment. Please feel free to contact me for any further information.

Sincerely,

Arie Kim, Ph.D.
Assistant Research Scientist



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To Whom It May Concern

Moscow

January 16, 2025

№ _____
Re: № _____

I am writing this letter of reference for Varvara Lazarenko, who has graduated from a Bachelor of the Faculty of Biology of Lomonosov Moscow State University in June, 2020.

I have known Varvara Lazarenko since she came to our working group on physiology of blood circulation as a 3rd year bachelor student at the Department of Human and Animal Physiology to perform her graduation project on the contribution of TASK-1 potassium channels to the regulation of vascular tone in intact arteries of different organs in rat. In order to carry out these studies, Varvara mastered a wire myography technique - a method of registration of arterial contractile responses, including small arteries of heart and kidney. Moreover, Varvara successfully mastered real-time PCR technique to evaluate mRNA content of TASK-1 channels in arterial tissue.

She proved to be a very skillful experimenter, determined and initiative researcher by actively participating in discussion and interpretation of the obtained data as well as in development of the research strategy. Her exceptional analytical skills and extensive knowledge of physiology made it possible to successfully perform statistical analysis of the experimental data and write a comprehensive literature review. As a result, her project received the highest possible mark and was named as one of the best in her class. Therefore, Varvara possesses all the qualities needed to be a successful independent researcher and it was a pleasure to be her supervisor.

Varvara was granted by MSU scholarship for her academic and scientific achievements, which shows her high academic potential. She took part in Russian and international conferences, where she demonstrated her great communication skills. Regarding personal characteristics of Varvara, she is an approachable person and a genuine team player.

Therefore, I think that Varvara Lazarenko would fit for your program and I highly recommend her to you.

Sincerely,

Anastasia A. Shvetsova
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